Ex-ante prediction of membership leakage in on-policy self-distillation (OPSD): experimental design v1.1 (2026-10-26)

Basis of this version

- **Read in full:**
  - OP-Print (arXiv 2610.31616).
  - The official OPSD code (`siyan-zhao/OPSD`): `opsd_trainer.py`, `opsd_train.py`, `scripts/run_opsd_{1b,4b,8b}.sh` and the README.
  - arXiv 2505.11837.
  - Barron, Gnanapragasam and Tschannen, "A Fix for Softmax and Cross-Entropy" (arXiv 2610.31642): §§1, 3, 7 and 8 read closely.
- **Read at abstract and key-passage level:** 2602.15919, 2605.25819, 2505.18773, 2609.36331, and LT-IQR (arXiv 2411.05743).
- **Markers:**
  - **[unverified]**, **[inference]** and **[estimate]** mark statements that do not come from reading a primary source.
  - **[design choice]** marks pre-registered thresholds that are ours, not taken from the literature.
- v0 (`op-print-design.md`, Chinese) and v1 (`op-print-design-v1.md`, Chinese) are left unchanged.

Changelog

v1 → v1.1

1. **Language.** The document is now fully in English. Content is otherwise unchanged except for the items below.
2. **Numerically stable Δ (new §1.4).** Per-token KL is now computed in log space following Barron et al. (Eq. 9), in fp32, with an fp32 LM-head projection.
   - The KL direction was checked against OP-Print's Δ and against OPSD's β = 0 loss. Both are KL(teacher-context ‖ student-context), so no reversal is needed.
3. **ε for the log-ratio outcome.** The ad-hoc ε (1st percentile of Δ) is replaced by an ε calibrated to the *measured* noise floor of the exact scoring pipeline, using true-Δ = 0 runs. This is a new pre-H0 sanity check (§1.5). Winsorization is kept.
4. **New training-side ablation (H5).** The standard OPSD loss is compared with a fixed log-space loss.
   - This is crossed with the weight-synchronization path (bf16 merge/unmerge vs native LoRA serving), which separates the two drift channels.
   - Softmax Collapse is noted as a secondary concern.
5. **New risk.** H0/H1 can fail for purely numerical reasons.
6. A references list has been added (§9).

v0 → v1 (carried over)

1. **Correction of a factual error in v0.** v0 stated that OP-Print used full-parameter training. That was my inference, and it is very likely wrong.
   - OP-Print Appendix A states that all runs use the fixed teacher. In the OPSD code, `fixed_teacher=True` raises an error unless PEFT is enabled.
   - OP-Print Table 3 matches the official OPSD scripts item by item:
   - β = 0, λ = 1, T = 1.1, top-p 0.95, top-k 20;
   - max grad norm 0.1, lr 5e-6;
   - token clip 0.05, or 0.06 for 8B only.
   - Those scripts use **LoRA (r = 64, α = 128, all linear projections) at lr 5e-6**.
   - OP-Print is therefore very likely LoRA r = 64 at 5e-6 for 3 epochs (the scripts use 30 epochs). This is **[strong inference]**: the paper never mentions LoRA.
2. **Pre-H0 fixes:**
   - LR is selected by the fraction of θ0's Δ removed on the training set, never by attack metrics.
   - The clip rate of the max-grad-norm is logged.
   - bf16 merge/unmerge is handled explicitly.
   - The full-parameter control moves to Qwen3-0.6B, with a memory budget.
3. **H1 rewritten:**
   - log-ratio primary outcome with an ε/winsorization rule;
   - placebo-membership permutation null;
   - log T and θ0 leverage added as covariates;

    - 4 splits + complements;
    - explicit reliability threshold;
    - out-calibration before pooling;
    - FPC bias handling;
    - kill criterion "c adds nothing over θ0 leverage + loss".
4. **H0 statistics.** Paired bootstrap on the AUC difference, and paired DeLong for G vs G-Max.
5. **H2 made comparative.** Matched ε and matched utility, with δ = 1e-5. DP-OPD-004 / DP-IER-E is a conditional arm.
6. **H3** is a dose axis on the existing `jsd_token_clip`.
7. **Minor changes:**
    - K disambiguated.
    - KL computed in chunks.
    - H1′ uses common random numbers (CRN) plus a θ0-teacher arm.
    - The "tension" claim about 2505.11837 is withdrawn.
    - An LT-IQR-style ceiling is added.
    - Risk stratification is promoted.
8. **New code findings:**
    - OPSD ships `--top_k_loss`, so H4 needs no new code.
    - The token clip is element-wise over vocabulary entries (§0).

0. Facts pinned from the paper and the code

- **Signal.** $\Delta_t = \mathrm{KL}(p_\theta(\cdot \mid p_t(q,r), y<t) \,\|\, p_\theta(\cdot \mid p_s(q), y<t))$.
    - Both sides use the attacked model θ.
    - Rollouts are drawn as $y \sim p_\theta(\cdot \mid p_s(q))$.
    - p_t is the privileged (teacher) context containing the reference solution r. p_s is the plain (student) context.
- **Gray-box scores.**
    - $G = -(1/K)\Sigma_k (1/T_k)\Sigma_t \Delta_t$.
    - G-Max $= -(1/K)\Sigma_k \max_t \Delta_t$.
    - Δ is computed over 64-token chunks.
- **Meaning of K.** K is the number of full rollouts (default 32; Fig. 5a ablates the rollout count and the chunk size).
    - Chunks are 64-token segments *inside* each full rollout (up to 1024 tokens).
    - Algorithm 1 describes only the black-box attack.
- **Training loss (OPSD, β = 0).** `F.kl_div(student_logp, teacher_logp, log_target=True)` is the summand $p_k(\log p_k - \log q_k)$, i.e. KL(teacher ‖ student). This is the same direction as Δ.
    - The teacher is the adapter-disabled base plus the privileged context.
    - No `.float()` upcast occurs before `log_softmax`, so under `--torch_dtype bfloat16` the loss is very likely computed on bf16 logits. **[inference]** This depends on the `transformers` version.
- **`jsd_token_clip`.**
    - `jsd` has shape `[B,T,V]`. `clamp(max=clip)` is applied element-wise *before* summing over the vocabulary; the sum is then divided by the token count.
    - clip = 0 disables it.
    - The README confirms it is point-wise rather than a cap on the vocabulary-summed loss, and that the loss can be negative.
- **Reported metrics** are AUC, Acc and TPR@5%FPR only.
    - G-Max AUC: 0.594 (1.7B), 0.611 (4B), 0.587 (8B).

1. Pre-H0 fixes and sanity checks

1.1 Learning-rate selection

- **Removal metric.** $\mathrm{f_rm} = 1 - \mathrm{mean}_{x \in \mathrm{train}} \bar{\Delta}_\theta(x) / \bar{\Delta}_{\theta 0}(x)$, computed on fixed CRN rollouts (§3).
- **Grid.** LR ∈ {5e-6 (official anchor), 2e-5, 5e-5, 1e-4}, with r = 64 and 3 epochs.
- **Selection rule [design choice].** Take the smallest LR such that:
    - f_rm ∈ [0.2, 0.5], and
    - θ0's pass@1 drops by at most 2 points.
- **No attack metric is consulted at any point.**
- **Clip logging.** Log the fraction of steps where the pre-clip norm exceeds 0.1, and the median norm.

- If clipping is active close to 100% of the time, training is effectively normalized-gradient descent with LR as the only step-size knob. Report this.

1.2 bf16 merge/unmerge

What the code does (read): at each vLLM sync, it calls `merge_adapter()` → `load_weights` → `unmerge_adapter()`.

[inference] This has two consequences:

1. Rollouts are generated by bf16-rounded merged weights, so they differ slightly from the training policy.
2. The in-place bf16 add/subtract is not exactly invertible, so base weights can accumulate drift. Because the fixed teacher *is* the adapter-disabled base, **the teacher itself can drift**.

Mitigations:

- Score with the adapter *unmerged* (HF forward pass, bf16 base + adapter), or merge in fp32.
- Hash and compute the L^∞ distance of the base weights before and after training.
- If drift is non-zero, serve the adapter through vLLM's native LoRA (`enable_lora`) without merging.
- Report the rollout-policy vs training-policy mismatch.

1.3 Full-parameter control

- **1.7B full-parameter training does not fit in 16 GB.** AdamW's fp32 m and v alone take 1.7B × 8 B ≈ 13.6 GB.
- **Qwen3-0.6B full-parameter instead [estimate]:**

Component	Memory
bf16 weights	1.2 GB
bf16 gradients	1.2 GB
fp32 AdamW m, v	4.8 GB (8-bit Adam ≈ 1.2 GB)
Frozen teacher copy (OPSD's fixed teacher needs PEFT, so full-parameter training needs an explicit copy)	1.2 GB
Activations (checkpointed) + chunked logits	2–3 GB
vLLM at a 0.3 share	≈ 4.8 GB
Total	**fp32 Adam ≈ 15–16 GB (too tight); 8-bit Adam ≈ 12 GB (feasible). To be measured.**

- **Position.** OP-Print's target is very likely LoRA, so this control is optional. It is used only if H0 fails, to test whether LoRA capacity is the cause.

1.4 Numerically stable Δ computation (new; Barron et al., 2610.31642)

What the paper says (verified wording).

- *Softmax Cancellation* "occurs when student logits are close to teacher logits, and produces non-zero gradients even when the student and teacher logits are equal" (abstract).
- Standard implementations compute p = softmax(z_T) and q = softmax(z_S) separately and subtract them. This loses precision as z_S → z_T, and the gradient "plateau[s] at a non-zero noise floor" (Fig. 1b).
- The quoted magnitudes are **residual gradients at z_S = z_T in a teacher-initialized toy distillation** (a + b mod 53, T = 4, Adam ε = 1e-8; §7, Fig. 7): "$\sim 10^{-17}$" in fp64, "$\sim 10^{-8}$" in fp32 (a brief KL spike) and "$\sim 10^{-3}$" in bf16 (drives the student away from the teacher).
- **These are gradient residuals, not KL-value floors.** The paper reports no noise floor for the KL *value*. Carrying the concern over to the scored Δ is **our inference**, which is why the floor is measured directly (§1.5).

Why it matters here [inference].

- For members, training drives $p_\theta(\cdot|p_s)$ toward $p_\theta(\cdot|p_t)$. At most positions both contexts therefore sit in the near-equal regime, where any additive numerical floor is largest *relative* to the true Δ.
- A floor inflates Δ̄_in proportionally more than Δ̄_out, which compresses log(Δ̄_out/Δ̄_in) toward 0. **H0/H1 could therefore fail for numerical rather than substantive reasons.**
- A second, probably larger source is that **bf16 logits are quantized**.
  - bf16 has 8 significand bits, so the spacing is $2^{-4} = 0.0625$ for $|z| \in [8, 16)$ and 0.125 for $|z| \in [16, 32)$.
  - Logit differences between the two contexts are therefore resolved only to about 0.06–0.1 at typical magnitudes. **[estimate]** Qwen3's logit magnitudes are unverified.
  - The log-space formula cannot recover information lost before it.

Procedure:

1. Compute the final hidden states in bf16, then apply **the LM-head projection in fp32** (upcast the hidden state and the weight). Do the whole KL computation in fp32.

2. Per position, let z_T be the logits under the privileged context p_t and z_S the logits under the plain context p_s. Then:
   - p = softmax(z_T);
   - δ = z_S − z_T;
   - d_k = δ_k − Σ_i p_i δ_i;
   - s_k = d_k − log1p(Σ_i p_i expm1(d_i)) = log(q_k/p_k);
   - **Δ_t = KL(p‖q) = −Σ_k p_k s_k.**
3. **Direction check.** OP-Print defines Δ = KL(p_θ(·|p_t) ‖ p_θ(·|p_s)) = KL(p‖q) with p from the teacher context and q from the student context. This is exactly Barron's convention (p = softmax(z_T), s = log(q/p)). **No reversal is needed.**
   - The same holds for OPSD's β = 0 training loss and for the θ0-teacher arm of H1′ (where z_T comes from θ0 under p_t).
   - If a reverse-KL variant (β = 1) is ever scored, swap the roles: δ′ = z_T − z_S, centred under q, giving KL(q‖p) = −Σ_k q_k s′_k.
4. **Equivalent non-negative form (our algebra; to be checked against fp64 in §1.5).** Since Σ_i p_i d_i = 0, the definition reduces to KL(p‖q) = log1p(Σ_i p_i expm1(d_i)) = log1p(Σ_i p_i [expm1(d_i) − d_i]).
   - Every term in the last expression is $\geqslant 0$, so there is no sign cancellation.
   - Use a series for |d_i| < 1e-3.
   - Keep −Σ p s as the primary implementation and this form as a cross-check.
5. **Chunking.** Process 64-token position chunks: 64 × 151,936 × 4 B ≈ 39 MB per side, vs ≈ 0.62 GB per side for a full 1024-token sequence.
   - Two reductions over the vocabulary are needed (Σ p δ, then Σ p expm1(d)).
   - Vocabulary chunking with a two-pass reduction is available if memory is tight.
   - Privileged-prompt KV caches are reused across the K rollouts (≈ 115 KB/token for Qwen3-1.7B, i.e. ≈ 0.9 GB for an 8k-token r **[estimate]**).

1.5 Noise-floor sanity check and calibrated ε (new; replaces the 1st-percentile ε)

This runs before H0, on one pilot model.

Three tests, all with true Δ = 0:

- **N0 (arithmetic).** Feed identical logit tensors (z_S = z_T) to both the standard (bf16 `F.kl_div`, as in OPSD) and the fixed routine.
  - Expected: the fixed routine gives exactly 0.
  - The standard routine may also give 0 when both sides take the same code path, so N0 alone is not informative.
- **N0′ (accuracy).** Use z_S = z_T + η with η ~ N(0, σ²) for σ ∈ {1e-6, …, 1e-1}. Compare each routine and precision against an fp64 reference KL, and record the absolute error vs σ.
- **N1 (end-to-end pipeline floor).** Run the *exact* scoring pipeline (same batching, chunking, KV reuse, fp32 head) with the privileged context replaced by the plain context, left-padded with attention-masked tokens to the real p_t length and with adjusted `position_ids`.
  - The true Δ is mathematically 0, but both forward passes follow the real, length-dependent numerical paths.
  - Record the per-sample aggregates exactly as the attack computes them:
  - the mean over chunks (for G);
  - **the max over chunks (for G-Max). The max of noise is biased upward, so it needs its own floor.**

Calibrated ε [design choice]:

- ε_G = 99th percentile of the N1 per-sample mean.
- ε_GMax = 99th percentile of the N1 per-sample max.
- Floor the ε at these values; never set them below. Winsorization at the 1/99 percentiles of v is kept.

Gate [design choice]:

- If the N1 floor is not at least 10× below the median Δ̄_in of the pilot model, the numerics are suspect. Fix the pipeline (fp32 head, deterministic kernels, identical batch shapes) before H0.
- Report the floor for both the standard and the fixed pipeline. Their difference is itself a result.

2. Hypotheses

H0: replication

Setup. Qwen3-1.7B, LoRA r = 64, the LR from §1.1, the fixed Δ pipeline from §1.4.

Tests:

- G-Max AUC minus the best baseline's AUC, by **paired bootstrap**: resample samples, 2000 replicates. The "best baseline" is selected on a disjoint half of the samples, to avoid selection bias.
- G vs G-Max by **paired DeLong** on each model, plus the spread across the 8 models. OP-Print's "G-Max is better" is currently neither significant nor marginal.

Diagnostic. Also report H0 with the standard bf16 scoring routine. If AUC rises materially with the fixed routine, numerical noise was masking signal.

H1 (main line): does θ0 concentration predict leakage?

Hypothesis. A concentration statistic c(x) computed under θ0 predicts the per-sample leakage v(x). For c we use the top-10% chunk share of Δ_θ0 and max/mean.

Primary outcome. v(x) = log((Δ̄_out + ε) / (Δ̄_in + ε)), with ε calibrated in §1.5. The secondary outcome is Δ̄_out − Δ̄_in. The G-Max version is a secondary analysis.

Calibration before pooling:

- Divide each model's Δ̄ by that model's median Δ̄ over its out-samples (multiplicative out-calibration). This keeps values positive, so the ratio stays meaningful.
- Per-model z-scores are used only for the difference outcome and for pooled ROCs. z-scoring would break the ratio.

Covariates d:

- θ0 pass@k;
- θ0's Δ̄;
- log T (mean θ0 rollout length);
- θ0's loss;
- θ0 last-layer leverage. This is the 2602.15919 method transplanted to θ0 hidden states; its transfer is **[unverified]**.

Placebo-membership null:

- For each sample, randomly relabel 4 of its 8 models as "in" (counts fixed), recompute v_placebo and refit. Repeat 200 times.
- c's coefficient on v must exceed the 95th percentile of the placebo distribution.
- c must not predict v_placebo.

Kill. Given d, c's partial-coefficient CI includes 0, or its incremental R^2 does not exceed the placebo. In words: **c adds nothing over θ0 leverage + loss.**

H1′: reference-calibrated attacks

Use **CRN**: Δ_θ and Δ_θ0 are computed on the same rollouts and the same chunks, with the §1.4 pipeline. Three scores:

1. S_θ − S_θ0.
2. The remaining correction ratio ρ(x) = ΣΔ_θ / ΣΔ_θ0, the per-sample analogue of ρ_S.
3. **The θ0-teacher arm**, KL(p_θ0(·|p_t) ‖ p_θ(·|p_s)), which is the training objective itself. A variant applies the clip.

Cost and novelty:

- All three scores need only the public base model and r.
- None appears in OP-Print.
- The extra cost is one θ0 prefill.

H2: comparative DP

Comparison. Two mechanisms at matched ε (δ = 1e-5 ≪ 1/N = 2e-3) and matched utility (held-out pass@1 within 1 point). The outcome is the OP-Print advantage, measured as TPR@1%FPR and AUC − 0.5 on the out-calibrated pooled ROC.

Arm A: DP-SGD.

- Per-record clipping on (q, r): all rollout gradients of q are summed into one before clipping.
- Poisson subsampling, PRV accountant.

Arm B: DP-OPD-004 / DP-IER-E.

- **Alex must confirm that its privacy unit is the distillation record (q, r)**, not the teacher's training data.
- Until confirmed, this arm is out of scope and is replaced by **arm B′**: DP-SGD with the token clip on, vs arm A with the clip off.

Prediction (falsifiable). B or B′ has a significantly lower OP-Print advantage than A, or there is no difference. Both directions are pre-registered.

"ε̂ ≪ ε" is no longer a conclusion, because it is the default outcome.

H3: dose response

Dose axis. `jsd_token_clip` ∈ {0 (off), 0.2, 0.05 (paper), 0.01}.

Prediction. Clipping acts element-wise at high-discrepancy entries, so the G-Max advantage should fall monotonically with dose.

Adaptive attacks (required):

- G (sum);
- the top-q chunk mean;
- a learned aggregator: logistic regression on the chunk-Δ vector, trained on 2 models and tested on 6.

Reporting and baseline. Report utility at each dose. DuoLearn (token-level empirical defense) is the comparator.

H4: top-k teacher (lowest priority)

OPSD's `--top_k_loss` ∈ {0, 20, 100} vs TT-OPD-style unbiased top-k, compared at matched utility.

H5: training-side numerical ablation (new)

Motivation:

- Barron et al. Fig. 7: with standard soft-target cross-entropy, a student initialized at the teacher is driven away from it in bf16 ("amplifies rounding error into parameter drift in bf16 and a brief KL spike in fp32"). The fixed loss stays at KL = 0.
- The mechanism: Adam normalizes any residual gradient that exceeds its ε into steps of size η.
- In OPSD there are two distinct drift channels **[inference]**:
  1. **Adapter drift.** The cancellation pseudo-gradient moves the LoRA adapter, i.e. the student, at positions where the true Δ ≈ 0. These are disproportionately member positions late in training.
  2. **Teacher drift.** Base-weight drift from bf16 merge/unmerge (§1.2) moves the adapter-disabled base, which is the fixed teacher.
- Both blur the in/out contrast that OP-Print exploits. The direction of their effect on AUC is not obvious, so the ablation could *raise* measured leakage.
- **Caveat [inference].** Barron's regime has a true gradient of exactly 0. In OPSD the aggregate gradient is dominated by high-Δ tokens and globally clipped at 0.1, so the effect size could be negligible. That is what H5 tests.

Design (2 × 2):

Factor	Levels
Loss	(a) standard OPSD bf16 `F.kl_div`; (b) **fixed**: a custom autograd function with forward −Σ p s from §1.4, and a backward that uses Barron's log-space u_k = p_k expm1(s_k) = q_k − p_k with the zero-sum correction g_k = u_k − q_k Σ_i u_i
Weight sync	bf16 merge/unmerge vs vLLM native LoRA

- **Clip-aware gradient (our derivation).** With the element-wise clip and unclipped set U, the gradient is g_j = 1j∈U − q_j(1 − P_U) + q_j·1[j∉U], where P_U = Σ_{k∈U} p_k.
  - Compute (q_j − p_j) via u_j, and 1 − P_U as the sum of the clipped p_k.
  - This must be checked against autograd in fp64 before use.
- **Scale.** Start with one split + complement pair per cell (2 models × 4 cells), on identical splits. Run all 8 splits only if a cell differs by more than the pre-registered threshold (|ΔAUC| ⩾ 0.02, paired bootstrap CI excluding 0 **[design choice]**).
- **Outcomes:**
  - OP-Print AUC, TPR@1%FPR and TPR@5%FPR, scored with the fixed pipeline in every cell;
  - base-weight L∞ drift;
  - held-out teacher–student KL on out-samples;
  - pass@1.

Secondary: Softmax Collapse.

- This affects high-confidence tokens: a margin above the "precision cliff" (≈ 5.5 in bf16, 16.6 in fp32) zeroes the target gradient (Barron et al. §2). In soft-target terms, p or q rounds toward one-hot.
- Such tokens are common in LLM rollouts but mostly have low Δ. **[inference]** They mainly affect G's mean and the noise floor, not G-Max's high-discrepancy positions.
- fp32 scoring plus log-space evaluation also mitigates them. No dedicated arm.

3. Setup and compute

Pool and design:

- OpenThoughts math subset, $N^+ = 1000$.
- **4 random splits + their complements = 8 models**, each trained on 500 samples. Every sample is exactly 4 in / 4 out.

Split-half reliability:

- Each half takes 2 (split, complement) pairs, so each half is exactly 2 in / 2 out. There are 3 such partitions; average over them.
- **Threshold:** raw split-half $r \geqslant 0.3$, which corresponds to ≈ 0.46 for the 8-model estimate after Spearman–Brown.

FPC bias (2605.25819, verified):

- Drawing $N = 0.5 \cdot N^+$ from a fixed superset gives FPC $= 1 - N/N^+ = 0.5$. In/out variance is underestimated and vulnerability is overestimated.
- **The complement design pins N/N^+ at exactly 0.5**, which is the worst case for this bias, so two critique points conflict here.
- **Resolution:** keep the complement design for H1, which concerns relative ranking, where FPC is approximately a common factor. Apply the paper's FPC correction wherever absolute TPR is reported. The details of its "Decomposed FPC" were only partially read **[unverified]**.
- Concatenated ROCs give inconsistent per-sample FPRs, so report out-calibrated results and note this.

Scoring:

- **$K = 8$ full rollouts.** Fig. 5a (4B) shows G-Max AUC ≈ 0.02 lower at $K = 8$ than at $K = 32$.
- Check rank stability with $K = 32$ on a 200-sample subset.
- "First 64 tokens only" is a cheap ablation. It is **not** the method: the paper's max is over the whole rollout.

Compute [rough estimate; calibrate on one model]:

Item	Estimate
Training, 8 models	$\approx$ 8–16 GPU·h
Scoring ($\approx$ 64k rollouts + two prefills)	$\approx$ 10 h; the fp32 head adds an unmeasured overhead
H1′ θ_0 pass	$\approx$ +3 h
LR grid	$\approx$ 4–8 h
H5 (4 cells × 2 models + scoring)	$\approx$ 8–12 h

LT-IQR-style training-time ceiling:

- Log each in-sample's Δ loss every time it is trained on, plus CRN Δ at `save_steps` checkpoints.
- Use the IQR ranking as the ceiling for information available during training, and compare it with c (θ_0, pre-training).
- **Caveat:** with 3 epochs, only 3 points per sample come for free. Checkpoint evaluations cost extra.

4. Prior work (verification status)

- **2602.15919** (Dorseuil, Atif, Cappé), verified.
  - Per-sample risk from a *single trained* model via loss plus last-layer leverage. **Not ex ante.**
  - Experiments cover classification and regression only (CIFAR, BloodMNIST, SST-2 BERT, tabular data). No generative LLMs.
  - Applying leverage to θ_0 is our adaptation **[unverified transfer]**.
- **LT-IQR** (Pollock et al., USENIX Security '25; arXiv 2411.05743), verified.
  - IQR of per-sample, per-epoch loss traces, with no extra models. 92% precision@1% on CIFAR.
  - Image classification only **[abstract]**.
- **2605.25819** (Jälkö et al.), verified:
  1. Shadow sets drawn from a fixed superset inflate vulnerability (FPC).
  2. Concatenated TPR is miscalibrated across per-sample FPRs.
  3. Fixes are proposed for both.
- **2505.18773** (Hayes et al., NeurIPS '25), verified.
  - Many per-sample decisions are coin-flip-unstable across training seeds.
  - Vulnerability correlates with sample length (§6.1, App. F).
- **2609.36331** (Rita, Zhang, Tramèr, AISec '26), verified.

  - Per-example calibration without reference models is "the common bottleneck" for one-round attacks and one-run DP auditing. Neighbour-based calibration is proposed.
  - Image tasks only. Relevant if H2 uses one-run auditing.
- **2505.11837.** The v0 claim of a "tension" with OP-Print is **withdrawn**. Its teacher–ground-truth KL on fixed data is a different quantity from OP-Print's teacher–student Δ on student rollouts.
- **DuoLearn (2502.19726).** H3 comparator.
- **2610.31642** (Barron et al.), verified: §1.4, §1.5 and H5.
  - All of its experiments are small grokking or toy-distillation models. Its conclusion itself says the regimes probed "are uncommon in standard large-scale training".
  - Transfer to LLM-scale OPSD is **[unverified]**, which is why the effect is measured, not assumed.

5. Execution order and stopping rules

1. Bring up the OPSD code. Then:
   - bf16 base-drift check (§1.2);
   - **noise-floor check N0/N0′/N1 and ε calibration (§1.5)**;
   - LR grid (§1.1; f_rm and pass@1 only).
2. **H0** on one model. **Kill:** the paired-bootstrap AUC-difference CI includes 0.
   - If H0 fails, first run the standard vs fixed scoring diagnostic.
   - Optionally run the 0.6B full-parameter control.
3. Train 8 models and apply the reliability gate. If **raw r < 0.3**, drop per-sample H1 and switch to risk stratification (§6).
4. H1 and H1′ in parallel, then the LT-IQR ceiling.
5. Then, in this order:
   - H3 (dose axis; no DP; cheap);
   - H5 (numerical ablation; can share models with H3's clip = 0.05 cell);
   - H2 (pending Alex's confirmation of arm B's privacy unit);
   - H4.

6. Risks

- **Risk stratification may be the main deliverable,** not merely a fallback, given 2505.18773's per-sample instability.
  - Bin samples into quintiles of c and test monotonicity of the pooled TPR@1%FPR.
  - This is more robust than per-sample ranking and still delivers an ex-ante risk indicator.
- **Numerical failure mode (new).** bf16 logit quantization, the softmax-cancellation floor and the inflated G-Max noise maximum can compress the in/out log-ratio. H0/H1 could fail for numerical rather than substantive reasons. §1.4–1.5 and the H0 diagnostic address this.
- The effect is weak (AUC $\approx$ 0.6).
- If the LoRA-configuration inference is wrong, replication may fail.
- bf16 drift may contaminate the teacher.
- Long r makes prefill costly.
- The sampling temperature must stay at T = 1.1.
- If arm B's privacy unit is not the distillation record, H2 reduces to A vs B′.

7. Where I disagree with the critique (unchanged from v1)

1. **"LoRA at 5e-6 barely moves the adapter."** The premise largely fails. The official OPSD configuration is LoRA r = 64 at 5e-6; OP-Print very likely used it and measured AUC 0.59 there. The Δ-removal LR procedure is still adopted because it is sound, with 5e-6 kept as the anchor.
2. **K and 64-token truncation.** K counts rollouts, and chunks live inside full rollouts, so truncation is an ablation only.
3. **The complement design conflicts with 2605.25819's FPC point.** Resolution in §3.
4. **Per-model z-scoring is incompatible with a ratio outcome.** Multiplicative out-calibration is used instead.
5. **The 0.6B full-parameter control is optional** (reason in §1.3).
6. **LT-IQR is not free in OPSD.** 3 epochs give 3 points per sample.

All other points were adopted.

8. Notes on the v1.1 additions (precision of claims)

- **Barron's $\sim 10^{-3}$ (bf16) and $\sim 10^{-8}$ (fp32) are residual gradient magnitudes at z_S = z_T in one toy setting**, not KL-value floors and not LLM measurements. Our ε comes only from the N1 measurement.

- The KL direction matches without reversal (§1.4, step 3).
- The log1p-form KL and the clip-aware gradient are our derivations, to be validated against fp64.

9. References

- **[OP-Print]** Q. Li, X. Wang. arXiv 2610.31616 (2026).
- **[OPSD code]** github.com/siyan-zhao/OPSD (`opsd_trainer.py`, `opsd_train.py`, `scripts/`, README; read 2026-10-26).
- **[Barron+26]** J. T. Barron, D. Gnanapragasam, M. Tschannen. "A Fix for Softmax and Cross-Entropy". arXiv 2610.31642 (2026).
- **[DLC26]** Dorseuil, Atif, Cappé. arXiv 2602.15919 (2026).
- **[LT-IQR]** Pollock, Shilov, Dodd, de Montjoye. USENIX Security '25, pp. 5525–5544; arXiv 2411.05743.
- **[JPRH26]** Jälkö, Pradhan, Räisä, Honkela. arXiv 2605.25819 (2026).
- **[Hayes+25]** Hayes et al. arXiv 2505.18773 (NeurIPS 2025).
- **[RZT26]** Rita, Zhang, Tramèr. arXiv 2609.36331 (AISec '26).
- **[KD-MIA]** arXiv 2505.11837.
- **[DuoLearn]** arXiv 2502.19726.
- **[Steinke+23]** Steinke, Nasr, Jagielski. "Privacy Auditing with One (1) Training Run", NeurIPS 2023 (one-run auditing; not re-read for this version).